GinCol Lab · Aula

Student worksheet

Physical Education · 1st year of Bachillerato · Your body, your lab · Class 1 of 4 · With Isidora

How much do we move?

Class video: https://youtu.be/eMlHbYy2yh0

The code leads to the video.

Name: _____ Date: _____

Before you start

1. Without thinking about it too much, think back to last week. On how many days did you move for at least an hour? On a school day, how many hours do you spend on the internet for fun (social media, videos, games, music)? How many hours do you usually sleep on a weekday night? It's just for you: you don't need to show it to anyone.

2. **At the end of the class.** Go back to your three answers. Do you reach your hour, which is the minimum? How do your hours of internet and sleep compare with what people your age say and with what is recommended? Are you active, sedentary, or both at once?

While you watch the video

1. **Your bet.** Out of every 100 girls aged 17 and 18 who study in Spain, how many say they move for at least an hour every day? And out of every 100 boys?

2. Is the daily hour recommended at your age the ideal or the minimum? What does it mean that the WHO says "at least"?

3. **Bone.** Why do people talk about a "window" in the years of the growth spurt? Why do impacts strengthen bone? Explain what happens inside the bone.

4. **The heart.** Why can the heart be trained? What changes in a trained heart?

5. **Muscle and mind.** What does muscle do with the sugar in the blood when it moves? What can moving do for your mood, and what can it not replace?

6. In the experiment with three weeks in bed, what happened to those five men? And what happened when they started training again?

7. Why is the survey figure (seven girls and nineteen boys in every hundred) not an exact measurement? Give two reasons. Why can we still trust the differences between boys and girls?

8. Why do we say that, with age, girls drop "in proportion" much more than boys?

9. **What eats up your day.** How many hours do students your age say they spend on the internet for fun? What counts as "internet for fun"? Where do Canada and Australia draw the line, and where does the Spanish Association of Paediatrics? How are they different?

10. How many hours of sleep do you need at your age? Why do paediatricians advise leaving your phone out of the bedroom?

11. Can someone be active and sedentary at the same time? Why?

12. Where can you find your hour? Why is Physical Education class not enough on its own?

The talk test

Try it yourself, safely and at your own pace: while you do each activity, try to talk and then to sing a song you know. If anything bothers you, stop.

Activity	Can you talk?	Can you sing?	Light, moderate or vigorous?
Strolling calmly			
Walking fast			
Going up some stairs at a good pace			
Skipping, or small hops on the spot, for one minute (if you can)			
Cycling to school (think about it: how do you ride?)			
The activity you do most in your free time:			

Does your bone feel it?

Half a minute for each movement, on a floor that isn't slippery. If anything hurts or you can't jump, do only the second row, or think it through.

Movement	Do you feel a jolt when you land or put your weight down?	Does it ask anything of your bone? Why?
Gentle little hops on the spot, landing quietly with your knees slightly bent		
Rising onto your toes and lowering again, slowly, without leaving the floor		
Think about swimming or cycling: where would the jolt be?		

Fact or explanation?

Mark **F** if it's a fact (the data say so) or **E** if it's an explanation that is still a hypothesis.

1. Girls aged 17 and 18 say they move less than boys their age.

2. Girls move less because they get less support from their family and their school.

3. Students aged 14 to 18 say they spend about five hours a day on the internet for fun, on weekdays.

4. Having your phone in your bedroom at night takes hours of sleep away from you.

5. In an experiment with five 20-year-old men, their aerobic capacity fell more in three weeks in bed than in the thirty years of ageing that followed.

6. People who feel less capable in Physical Education tend to move less.

Practice

1. **Lucía's week** (made up): Monday 20, Tuesday 70, Wednesday 45, Thursday 90, Friday 30, Saturday 120 and Sunday 60 minutes of moderate or vigorous activity. Check whether she reaches the WHO average (at least 60 minutes a day, on average over the week). What would she answer to the survey question: on how many days did she move for at least 60 minutes?

2. Explain in your own words why skipping asks more of your bones than swimming, and why both are good for your heart.

3. **Your week.** For seven days, write down each day your minutes of moderate or vigorous activity (use the talk test to decide which ones count), your hours of internet for fun and your hours of sleep. Work out the averages and compare them with what you saw in the class: do you move enough?

	Mon	Tue	Wed	Thu	Fri	Sat	Sun	7-day average	Monday–Friday average
Minutes of moderate or vigorous activity									
Hours of internet for fun									
Hours of sleep (the night before)									

To compare yourself: does your average number of minutes reach 60? On how many days did you reach 60 (what you would answer in the survey)? How do your Monday-to-Friday internet hours compare with the roughly five that people your age report, and with the two-hour line? Does your Monday-to-Friday sleep reach 8 hours? Your numbers are yours: nobody has to see them.

Keep thinking

Why do you think that, with age, the girls who say they move every day drop, in proportion, much more than the boys? And what would you change so that it didn't happen?

The data in the class come from the HBSC 2022 study and the ESTUDES 2025 survey (Spain's Ministry of Health), from the World Health Organization (2020) and from the studies cited in the teacher sheet. Lucía's week is made up.

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